

Date	15 March 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details

S.No	Architecture Component/Layer	Technology Chosen	Justification / Purpose
1	Frontend/ Client-Side	React.js & Tailwind CSS	React provides a highly responsive, component-based UI perfect for a banking dashboard. Tailwind CSS allows for rapid, consistent styling of the interface for bank agents.
2	Backend/ Server-Side	Python with FastAPI	Python is the industry standard for machine learning. FastAPI is extremely fast and perfect for creating the REST APIs needed to serve the ML prediction model to the frontend in real-time.
3	Database / Data Storage	PostgreSQL	A robust, relational database is ideal for securely handling structured financial data, applicant records, and storing historical prediction logs.

4	Cloud/Hosting/ Deployment	Docker & AWS (EC2/ S3)	Docker ensures the ML model and app run consistently across environments. AWS provides scalable hosting (EC2 for the app) and secure storage (S3 for raw datasets and model artifacts).
S.No	Architecture Component/Layer	Technology Chosen	Justification / Purpose
5	Version Control & CI/ CD	GitHub & GitHub Actions	GitHub ensures reliable version control of the code and ML pipelines. GitHub Actions enables automated testing and deployment whenever the model or code is updated.
6	Third-Party APIs / Other Tools	Scikit-Learn, XGBoost, & SHAP	Scikit-Learn and XGBoost are the core ML libraries used to train the classification model. SHAP (SHapley Additive exPlanations) is used to generate the explainability reports for each approval decision.